MILITARY SPECIFICATION

BALLISTICS COMPUTER SUBASSEMBLY: 8635273 (CONTROL UNIT)

1. SCOPE

1.1 This specification covers one type of electrical mechanical assembly known as Ballistics Computer Subassembly: 8635273, hereinafter referenced as "Control Unit" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–F–13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL–P–14232 —Parts Equipment and Tools for Ordnance Materiel: Packaging of
MIL–I–45208 —Inspection System Requirements
MIL–I–45607 —Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes
MIL–STD–109 —Quality Assurance Terms and Definitions
MIL–STD–202 —Test Methods for Electronic and Electrical Component Parts

DRAWING

U.S. ARMY MUNITIONS COMMAND

F8635273—Ballistics Computer Subassembly: 8635273 (Control Unit)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. The control unit shall be manufactured in accordance with Drawing F8635273, and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.3 General specification. The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL–F–13926:

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(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.4 Environmental.

3.4.1 *Operating temperatures.* The control unit shall meet all the requirements of 3.6 while exposed and thermally stabilized at ambient temperatures of —40° and +150° Fahrenheit (F). Upon return to standard ambient temperature the control unit shall meet the requirements of 3.4.2.

3.4.2 *Vibration.* The control unit shall meet the requirements of 3.5 to 3.6.2 inclusive after being vibrated in a vertical plane for five minutes plus or minus 15 seconds at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second (cps).

3.4.3 *Water-resistance (sealing).* The interior of the control unit shall show no evidence of water penetration or other damage after being subjected to a simulated rainfall for a time duration of 15 minutes ± 15 seconds.

3.5 Electrical.

3.5.1 *Continuity.* The continuity of the control unit shall conform to Electrical Schematic Drawing C8635249.

3.5.2 *Dielectric strength.* There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the control unit are subjected to an alternating current (ac) root mean square (rms) potential of 300 volts plus twice the rated voltage at a frequency of 60 cps for a period of not more than 30 seconds nor less than five seconds.

3.5.3 *Insulation resistance.* All insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 10 megohms with 500 volts direct current (vdc) applied.

3.6 Performance.

3.6.1 *Switch operation.* With a 24 vdc 6.5 amp resistive load connected in series with each switch, the maximum voltage drop through each respective pin set, F and J, F and K, F and L and F and M shall not exceed 0.300 volts.

3.6.2 *Switch lens illumination.* The application of positive and negative leads of a 24 vdc source connected to pins E and F respectively shall illuminate the four switch lenses with equal intensity from dim to full intensity when the rheostat is rotated from the counterclockwise to the clockwise stop. Movement of one additional positive lead from pins A to B, B to C and C to D shall increase the intensity of the red, blue, yellow and green lenses respectively.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the government as specified in the contract or purchase order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services that conform to prescribed requirements.

4.4.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL–I–45208 and the component inspection requirements of Specification MIL–F–13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL–STD–109.

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4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in Standard MIL–STD–105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL–F–13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL–STD–105, and the inspection level and sampling plan specified in table I titled "Classification of Defects". Examinations and tests for packaging, packing and marking shall be in accordance with Specification MIL–P–14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement and to reject individual non-conforming items.

TABLE I. *Classification of defects*

Use Level II of Table I with
Sampling Plan Table II-A of Standard MIL–STD–105

	Requirements	*Test procedure*
Critical: None		
Major: AQL 1.0%		
101. Physical failure, post vibration	3.4.2	4.5.1.2
102. Continuity	3.5.1	4.5.2
103. Switch operation	3.6.1	4.5.5.1
104. Switch lens illumination	3.6.2	4.5.5.2
Minor: None		

Note. The above tests shall be conducted at standard ambient temperature 60° to 90°F.

4.2.3 *Disposition of non-conforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL–STD–105.

4.3 Control tests.

4.3.1 *General.* One control unit shall be selected at random from each one hundred produced, or from each month's production, whichever occurs first. The control sample shall meet the following requirements and tests:

	Control test	*Requirement*	*Test procedure*
301.	Materials	3.1	as per spec. or dwg.
302.	Fabrication	3.2	visual—per dwg.
303.	General specification	3.3	visual—MIL–F–13926

4.3.2 *Environmental and electrical.* Three control units shall be selected at random from each fifty produced, or from each month's production, whichever occurs first. The control sample shall meet the following requirements and tests:

	Control test	*Requirement*	*Test procedure*
304.	Operating temperature	3.4.1	4.5.1.1
305.	Water resistance	3.4.3	4.5.1.3
306.	Dielectric strength	3.5.2	4.5.3
307.	Insulation resistance	3.5.3	4.5.4

4.3.3 *Control test failure.* Should any one item of the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirement of Specification MIL–I–45208 and all applicable requirements specified in Specification MIL–I–45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, con-

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struction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.5 Test methods and procedures.

4.5.1 *Environmental.*

4.5.1.1 *Operating temperatures.* This test shall be conducted in conformance with the temperature specified in 3.4.1. The test equipment shall be in accordance with the "Test Facilities" requirements specified in Specification MIL–F–13926. While thermally stabilized at the specified temperatures, the control unit shall be subjected to the tests specified in 4.5.5 to 4.5.5.2 inclusive to determine conformance with the requirements referenced therein.

4.5.1.2 *Vibration.* To determine conformance to 3.4.2, the control unit shall be mounted in its normal position using a suitable mounting apparatus to attach it to a vibrating machine or fixture. Test equipment shall be in accordance with the "Test Facilities" portion of Specification MIL–F–13926 and shall be capable of providing vibration of the amplitude and frequency for the time specified in 3.4.2. After vibration, the control unit shall be given a visual-tactile examination for damage and loose parts. The control unit shall then be subjected to the tests specified in 4.5.2 to 4.5.5.2 inclusive to determine conformance with the respective requirements referenced therein.

4.5.1.3 *Water resistance.* To determine conformance to 3.4.3, this test shall be conducted in accordance with Procedure I of Specification MIL–F–13926 except for the time duration specified in 3.4.3. The control unit shall show no evidence of water penetration as a result of this test when visually examined.

4.5.2 *Continuity.* The control unit shall be tested, using standard measuring equipment, to demonstrate continuity of the circuits in accordance with Schematic Drawing D8635307 to determine compliance with 3.5.1.

4.5.3 *Dielectric strength.* The control unit shall be tested in accordance with Method 301 of Standard MIL–STD–202 and shall meet the requirements specified in 3.5.2.

4.5.4 *Insulation resistance.* Immediately after satisfactorily withstanding the test for dielectric strength, the control unit shall be given the insulation resistance test in accordance with Method 302 of Standard MIL–STD–102 (test condition B) and shall meet the requirements specified in 3.5.3.

4.5.5 *Performance.* These tests shall be conducted using a 24 vdc power supply, voltmeters connectors, leads, and a mounting bracket, all connected into an electrical network with the 6.5 ampere resistive load specified. Test equipment accuracy shall be in accordance with 4.4.1.

4.5.5.1 *Switch operation.* To determine acceptable switch operation, the control unit shall be connected to the applicable equipment and circuit specified in 3.6.1 and 4.5.5. With proper voltage applied, each switch shall be depressed, and the voltage drop across each set of pins shall be as specified in 3.6.1.

4.5.5.2 *Switch illumination.* This test shall be performed with the control unit connected to the DC power supply as specified in 3.6.2. With the dimmer control at the extreme clockwise position, note the brightness of the switch lenses. Advance the dimmer slowly to its extreme counterclockwise position and at the same time note the decreased light intensity as the range of the dimmer is traversed. With an additional positive voltage lead applied to each lamp pin connection of J101 the resulting increase in light intensity shall be noted. The switch lenses shall illuminate as specified in 3.6.2.

5. PREPARATION FOR DELIVERY

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5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Specification MIL–P–14232 and the applicable packaging data sheet. The level of protection shall be as specified in the procurement document (see 6.2).

6. NOTES

6.1 Intended use. This control unit is primarily intended as a functional component of the electronic analog Computer Ballistic: M16. A choice of any one of four ammunitions can be made from the control unit. The choice of ammunition is made by depressing a color coded control button. An increase in the illumination of the selected button indicates which ammunition is selected inside the computer unit.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and data of this specification.
(b) Selection of an applicable level of preservation, packaging and packing (see 5.1).
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

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